

MANAS MALLIK

MECHANICAL DESIGN ENGINEER

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PROFESSIONAL SUMMARY

Mechanical Design Engineer with over one year of experience in full-cycle product development and mechanical packaging for defense-grade systems. Expertise in 3D CAD modeling (SolidWorks, Fusion 360, Creo), sheet metal and plastic part design, DFM/GD&T implementation, and detailed BOM and technical documentation. Proven ability to deliver robust assemblies and prototypes for complex sensor and UAV applications in global engineering teams.

CORE COMPETENCIES

3D CAD Modeling & Tools: SolidWorks (Advanced Part Modeling, Complete Assemblies, Motion Study, Drafting), Fusion 360, AutoCAD, Creo (Basics), 3D Modeling, Parametric Detailing
Product Design & Drafting: Mechanical Product Design, Assembly Design, Enclosure Configurations, 2D Detailed Engineering Drawings, Generative Layout Detailing, Mechanical Packaging Design
Specialized Manufacturing: Design for Manufacturing (DFM), Sheet Metal Design, Plastic Part Design, Geometric Dimensioning & Tolerancing (GD&T), Injection Molding Layouts, Metal Parts Machining, Sheet Metal Process, Casting, Welding, Metals
Validation & Documentation: Bill of Materials (BOM) Preparation, Prototype Development, Technical Documentation, Change Request Engineering, Six Sigma Methodology
Simulation & Analysis: ANSYS Fluent (CFD), Finite Element Analysis (Basic FEA), Shockwave/Pressure Profile Modeling

PROFESSIONAL EXPERIENCE

IG Defence

Engineer 1 - Mechanical Design

Present

Delhi, India

- Design, map, and execute robust 3d modeling and mechanical packaging design for structural assemblies and sheet metal enclosures for defense-grade UAV systems.
- Enforce strict design for manufacturing limits and GD&T principles to achieve assembly precision, material optimization, and production viability in sheet metal process workflows.
- Deliver exhaustive manufacturing packages, including detailed 2D engineering drawings, exact product layout specifications, and comprehensive Bills of Materials (BOM).
- Develop prototypes utilizing metal parts machining and welding techniques, and conduct verification testing in compliance with product requirements.

Defence Research & Development Organisation (DRDO)

Summer Intern

May 2024 - June 2024

Chandipur, Odisha

- Engineered complex 3d modeling profiles with parametric detailing and investigated high-displacement shockwave pressure distributions for supersonic structures.
- Leveraged advanced CAD assemblies and analytical validation strategies to capture wave-pulse geometry and vector behaviors.

National Aluminium Company Limited (NALCO)

Industrial Trainee

June 2023

Angul, Odisha

- Interfaced with heavy industrial plant machinery layouts, applying Six Sigma methodology to assess and support operational equipment troubleshooting and workflow efficiency.

PROJECTS

SAEISS DDC 2025

2024 - 2025

Team Aerotech

- Led cross-functional CAD modeling efforts using SolidWorks for system architecture layout, rapid prototyping, and full assembly validation of a fixed-wing UAV
- Optimized material selection and conducted FEA-based analyses to secure component structural integrity while maximizing physical payload distribution
- Key Distinction: Captured All India Rank 1 (AIR 1) for Best Design Report at the national SAE DDC competition, proving exceptional technical reporting and validation skills.

Plastic Injection Hand Mold Design

June 2023

Central Tool Room & Training Centre (CTTC)

Bhubaneswar, Odisha

- Modeled and simulated tool-room injection molds in Creo, then generated production drawings for CNC toolpaths including milling, turning, and grinding
- Developed DFMEA and process flow charts to mitigate molding risks and ensure design for manufacturability with engineering plastics

EDUCATION

Veer Surendra Sai University of Technology (VSSUT)

B.Tech, Mechanical Engineering

2026

Burla, Odisha

- GPA: 8.02

Upendra Nath Higher Secondary School

Class XII

2020

Gandibed Nodal High School

Class X

2018

CERTIFICATIONS

- SolidWorks Industrial Certification: Central Tool Room & Training Centre (CTTC), Bhubaneswar
- ANSYS Fluent Virtual Internship (Basic FEA/CFD Integration): AICTE
- Computer Numerical Control (CNC Toolpath Programming): World Steel Association